

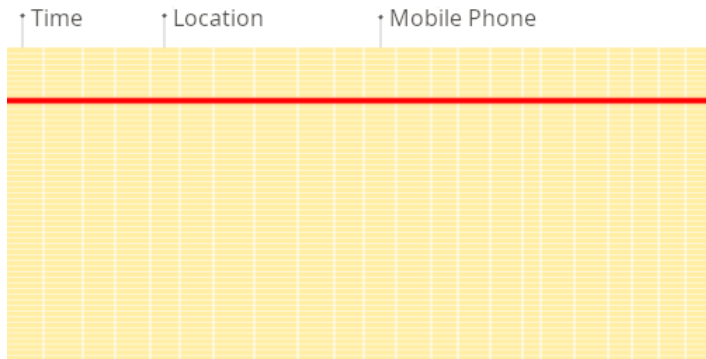


Tutorial 6: Column Oriented Databases

+ Question 1

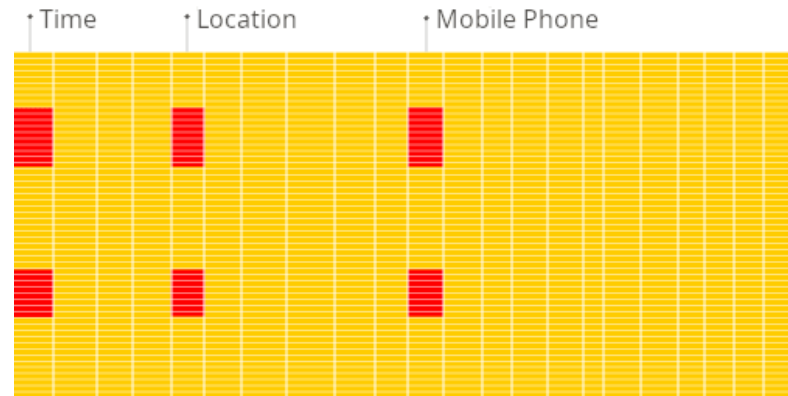
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- Prepare a column-oriented representation for the following row-oriented table.



A diagram of a row-oriented table. It consists of a grid of 30 columns and 10 rows. The columns are grouped into three sections: 'Time' (first 10 columns), 'Location' (middle 10 columns), and 'Mobile Phone' (last 10 columns). A single horizontal red line spans the entire width of the grid, representing a single row of data.

Row-oriented



A diagram of a column-oriented table. It consists of a grid of 30 columns and 10 rows. The columns are grouped into three sections: 'Time' (first 10 columns), 'Location' (middle 10 columns), and 'Mobile Phone' (last 10 columns). The data is organized by column, with red blocks indicating data for each column across the rows.

Column-oriented

+ Question 1

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- Prepare a column-oriented representation for the following row-oriented table.

TranID	TranDate	ProductCode	Category	CustNo	Qty	TotalAmt	Discount
7582	2023-08-01	GT4235	Media	V899	7	350.00	10%
7431	2023-08-01	HT8865	Food	N776	24	150.00	0%
9921	2023-08-01	RT5412	Gadget	V899	5	250.00	10%
4143	2023-08-02	GT1245	Media	N998	100	1200.00	0%
8876	2023-08-02	TR554	Food	V665	12	24.00	10%
9900	2023-08-02	GT4235	Media	V665	50	2500.00	10%
6548	2023-08-02	HT8865	Food	V665	24	150.00	10%

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- Prepare a column-oriented representation for the following row-oriented table.

The column-oriented table:

TranID: [7582, 7431, 9921, 4143, 8876, 9900, 6548]

TranDate: ['2023-08-01', '2023-08-01', '2023-08-01', '2023-08-02', '2023-08-02', '2023-08-02', '2023-08-02']

ProductCode: ['GT4235', 'HT8865', 'RT5412', 'GT1245', 'TR554', 'GT4235', 'HT8865']

Category: ['Media', 'Food', 'Gadget', 'Media', 'Food', 'Media', 'Food']

CustNo: ['V899', 'N776', 'V899', 'N998', 'V665', 'V665', 'V665']

Qty: [7, 24, 5, 100, 12, 50, 24]

TotalAmt: [350.00, 150.00, 250.00, 1200.00, 24.00, 2500.00, 150.00]

Discount: ['10%', '0%', '10%', '0%', '10%', '10%', '10%']

+ Question 1

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- Give an example of an SQL query that will perform 'significantly' better with the column-oriented representation.

TranID	TranDate	ProductCode	Category	CustNo	Qty	TotalAmt	Discount
7582	2023-08-01	GT4235	Media	V899	7	350.00	10%
7431	2023-08-01	HT8865	Food	N776	24	150.00	0%
9921	2023-08-01	RT5412	Gadget	V899	5	250.00	10%
4143	2023-08-02	GT1245	Media	N998	100	1200.00	0%
8876	2023-08-02	TR554	Food	V665	12	24.00	10%
9900	2023-08-02	GT4235	Media	V665	50	2500.00	10%
6548	2023-08-02	HT8865	Food	V665	24	150.00	10%

+ Question 1

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- Give an example of an SQL query that will perform 'significantly' better with the column-oriented representation.

Example: Calculate the total sales amount for each product category on a specific date.

```
SELECT Category, SUM(TotalAmt) AS TotalSales  
FROM Table  
WHERE TranDate = '2023-08-02'  
GROUP BY Category;
```

We only need few columns!

+ Question 1

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- Give an example of an SQL query that will perform 'significantly' better with the row-oriented representation.

```
SELECT *  
FROM YourRowOrientedTable  
WHERE TranID = 8876;
```

Column oriented database is beneficial when only a few columns are required!

+ Question 2

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- Match the following compressions for the most suitable column in the provided example in Q1:
- a) Dictionary Encoding
- b) Bit-Packing Encoding -T64
- c) Delta Encoding

+ Question 2

1

- Match the following compressions for the most suitable column in the provided example in Q1:

TranID	TranDate	ProductCode	Category	CustNo	Qty	TotalAmt	Discount
7582	2023-08-01	GT4235	Media	V899	7	350.00	10%
7431	2023-08-01	HT8865	Food	N776	24	150.00	0%
9921	2023-08-01	RT5412	Gadget	V899	5	250.00	10%
4143	2023-08-02	GT1245	Media	N998	100	1200.00	0%
8876	2023-08-02	TR554	Food	V665	12	24.00	10%
9900	2023-08-02	GT4235	Media	V665	50	2500.00	10%
6548	2023-08-02	HT8865	Food	V665	24	150.00	10%

TranID	TranDate	ProductCode	Category	CustNo	Qty	TotalAmt	Discount
Delta Encoding	Dictionary Encoding	Dictionary Encoding	Dictionary Encoding	Dictionary Encoding	Bit-Packing Encoding	Bit-packing Encoding	Delta Encoding / Bit-packing

+ Question 2

1

- Match the following compressions for the most suitable column in the provided example in Q1:

TranID	TranID
Delta Encoding	7582
	7431
	9921
	4143
	8876
	9900
	6548

The difference between each value in TranID is relatively smaller, so Delta Encoding is suitable.

+ Question 2

1

- Match the following compressions for the most suitable column in the provided example in Q1:

TranDate	ProductCode	Category	CustNo
Dictionary Encoding	Dictionary Encoding	Dictionary Encoding	Dictionary Encoding

For TranDate, ProductCode CustNo and Category, they are no numerical type data but there are multiple repeated values, thus dictionary encoding is preferred.

TranDate	ProductCode	Category	CustNo
2023-08-01	GT4235	Media	V899
2023-08-01	HT8865	Food	N776
2023-08-01	RT5412	Gadget	V899
2023-08-02	GT1245	Media	N998
2023-08-02	TR554	Food	V665
2023-08-02	GT4235	Media	V665
2023-08-02	HT8865	Food	V665

+ Question 2

1

- Match the following compressions for the most suitable column in the provided example in Q1:

Qty	Qty
Bit-Packing Encoding	7
	24
	5
	100
	12
	50
	24

For Qty, most of them are many small numbers, which can be transposed to compressed value by Bit-Packing Encoding.

+ Question 2

1

- Match the following compressions for the most suitable column in the provided example in Q1:

Discount	Discount
Delta Encoding / Bit- packing	10%
	0%
	10%
	0%
	10%
	10%
	10%

For Discount, if it can have floating values, delta encoding is preferred. If it only has integer values from 0 to 100, we can also use bit-packing encoding.

+ Question 2

1

- Match the following compressions for the most suitable column in the provided example in Q1:

TotalAmt	TotalAmt
Delta Encoding	350.00
	150.00
	250.00
	1200.00
	24.00
	2500.00
	150.00

For TotalAmt, because they are integers and continuous, it is suitable for delta encoding.

+ Question 3

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- Explain how query performance is optimized using:
 - a) Sparse Primary Index
 - b) Late Materialization
 - c) Projections

+ Question 3

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- Explain how query performance is optimized using:
 - a) Sparse Primary Index
- Unlike a regular primary index that covers every row in the table, a sparse primary index covers only a subset of rows.
- It divides the table into smaller partitions based on certain criteria (usually a range of values in a specific column). Each partition is then associated with an index entry, which points to the starting row of that partition.
- This technique is particularly useful when we have tables with a wide range of values in the indexed column, as it reduces the number of index entries and allows for quicker data retrieval for specific ranges.

+ Question 3

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- Explain how query performance is optimized using:
- b)Late Materialization
- Retrieve and process columnar format as long as possible before converting it into row format
- Query Processing: columns are filtered, projected and aggregated independently before combining. Row assembly (materialization) happens at final step.

+ Question 3

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- Explain how query performance is optimized using:
- b) Late Materialization
- Efficiency: reduces the amount of data loaded into memory by only accessing the necessary columns. Delays the combination of columns into rows, saving unnecessary computational cost.
- Comparison to Early Materialization: in early materialization, columns are combined into rows earlier in query processing, leading to inefficiencies.

+ Question 3

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- Explain how query performance is optimized using:
- c) Projections
- A Projection refers to creating an additional internal table synchronised with original table, such as filtered projection, aggregation, sorted projection.
- It improves query performance by selecting the projection that produces same correct result. This reduces the amount of data that needs to be read from disk or transferred over the network, leading to faster query performance.
- Creating and maintaining projections can require some overhead, especially if the original table is frequently updated.